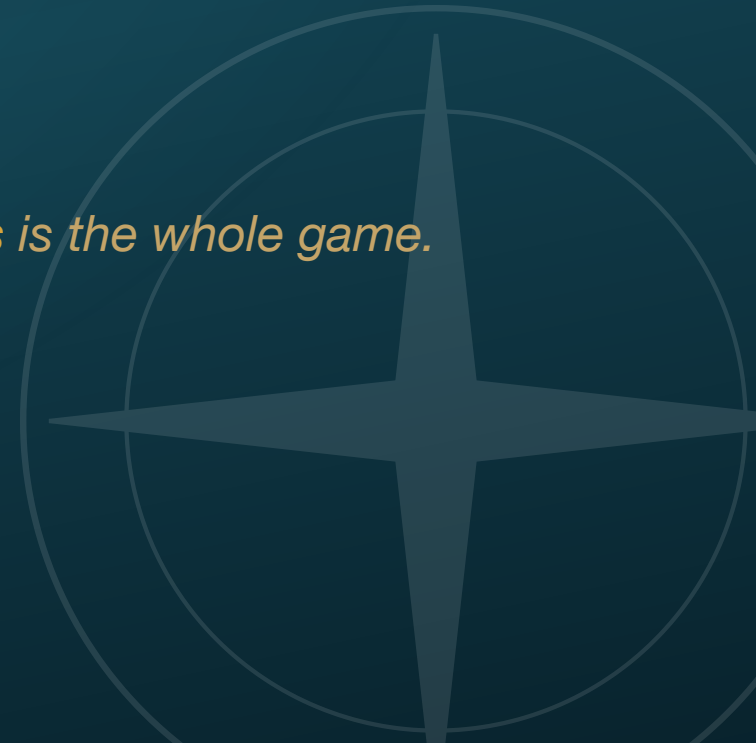


DAY 2 · EXPOSING AN APP · ~3 MINUTES

Gateway API

Who does what

*Three objects. Three different owners. Knowing which one is **yours** is the whole game.*



The old way put everyone on one wheel

- ♦ The classic way to expose an app was a single **Ingress** object.
- ♦ It mixed two jobs into one file: **run the front door** (the public IP, the open ports, the TLS certificate) *and* **route my app** (send this hostname to my Service).
- ♦ So the IT admin and the app developer were editing the **same object** — stepping on each other, with vendor-specific annotations bolted on.
- ♦ **Gateway API splits those two jobs into separate objects, owned by separate people.** That separation *is* the point.

Three objects, three owners

Object	Who owns it	What it decides
GatewayClass	Infrastructure provider	<i>What kind</i> of gateway exists (here: Traefik). Built once. You never see it.
Gateway	Cluster operator	The public IP, which ports are open (80/443), the TLS cert , and who may attach routes.
HTTPRoute	Application developer	"Traffic for my hostname → my Service." Lives in <i>your</i> namespace.

- ♦ One **Gateway** for the whole cluster. **Many** HTTPRoutes hang off it — one per app.
- ♦ Your Service stays **ClusterIP** (internal). The Gateway is the front door; your route is the sign that points visitors to your berth.

Who's who in *this* room

- ♦ **GatewayClass → the platform.** Traefik, installed with the cluster. Already done for you — invisible all week.
- ♦ **Gateway → the cluster admin.** This week that's *me*. On a real supported cluster at your college, that's your **IT department**: they stand up `main-gateway` once, wire the DNS, and load the wildcard TLS cert.
- ♦ **HTTPRoute → you.** In Lab 03 you write **one HTTPRoute** for your frontend — and *nothing else* about the gateway. You don't touch the IP, the ports, or the cert. You file a route; the front door was already built.

The takeaway

*IT builds the gateway **once**. Every developer just **files a route** against it.*

That clean split between operator and developer is the entire reason Gateway API exists.